# PlantCare : Machine Learning – Driven System for Optimized Growth, Watering, and Health Prediction of Plants

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***Abstract*—Plant diseases significantly impact crop yield, food availability and the economic stability of farming communities across the globe. Prompt and precise identification of these diseases is essential for preventing large-scale agricultural losses enabling effective intervention stratergies. Conventional diagnostic practices depend largely on visual inspection by experts, a process that labour-intensive, expensive and susceptivity. The work proposes an intelligent plant health monitoring and crop care recommendation platform powered by a convolutional neural network build upon the EfficientNet architecture. Through a web-based interference, user can upload images which are processed by the trained model to determine whether the plant is health or affected by disease, accompanied by a probabilistic confidence estimate. To increase the real-world usability, the framework incorporates location aware whether detection and generate water level based on plant and climate of the region. The complete solution is web based interactive solution developed using Gradio framework, enabling real-time prediction and seamless user prediction.**

***Keywords—Plant Disease Detection, Convolutional Neural Network, EfficientNet, Image Classification, Weather-Aware Agriculture, Gradio.***

## I. Introduction

Agriculture remains fundamental to global food system, human survival and economic development especially in regions heavily dependent on farming. Although modern cultivation practices have improved productivity, plant diseases persist as major constraint on agricultural output, negatively affecting both crop yield and quality. Detecting such diseases at early stage is vital to limiting widespread damages and enabling prompt corrective measures. Traditional diagnostic approaches rely largely on manual visual assessment y farmers or agronomists, making the process labour- intensive, expensive and prone to subjectivity. Moreover, symptoms in their initial stages are frequently inconspicuous, rendering accurate recognition by human eye particularly challenging.

Recent process in artificial intelligence and deep learning has accelerated the creation of automated pant disease diagnostic system based on digital imagery. Convolution Neural Networks (CNNs) have provided high effective for visual recognition tasks as they can autonomously extract hierarchical and discriminative patterns directly from raw Input images. Unlike convolution machine learning approaches that rely on handcrafted features, deep learning models offer superior representational capacity andimproved classification performance. Consequently, CNN- driven plant disease identification frameworks have attached considerable interest with precision agriculture and smart framing domains.

Beyond simply identifying diseases, good crop management also depends on environmental conditions such as temperature and humidity, which strongly influence how diseases spread and how much water crop need. However, menu existing plant disease detection system focus only on classification accuracy and do not consider real-time whether data or provide follow-up guidance after detection. This limits their usefulness in real farming situations, where growers need practical, easy-to-understand advice rather than just a disease label.

The study presents an intelligent plant health assessment and crop care advisory system that brings deep learning techniques with real-time information and agricultural expertise. Users interact with platform through a web-based interface by submitting images of plant leaves, which are analysed by trained CNN to identify the presence of disease and estimate prediction confidence. By considering both the diagnosed conditions and the user's geographical condition the system access live whether data to generate customised irrigation recommendations. In addition to disease identification, the framework provides condition-specific treatment guidelines along with direct link to suitable agricultural products, helping farmers to take timely and well-informed corrective actions.

The main goal of the proposed system is to create a practical, us er-friendly, and intelligent decision-support tool that helps farmers and agricultural professionals detect plant disease in early stage and manage crops more effectively. By integrating image-based disease identification with environmental context and clear, actionable recommendations, the system aims to improve overall crop health while reducing reliance on constant expert supervision.

## II. Literature Survey

Traditional plant disease detection systems relied heavily on classical image processing techniques combined with machine learning classifiers. These approaches extracted handcrafted features such as color histograms, texture descriptors, and shape-based characteristics from leaf images. The extracted features were then classified using algorithms like Support Vector Machines (SVM) and k-Nearest Neighbors (kNN). While these systems achieved reasonable accuracy on controlled datasets, they demonstrated poor generalization capability when exposed to real-world environmental variations such as fluctuating

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lighting conditions and background noise [1], [2]. With the advancement of deep learning, Convolutional Neural Networks (CNNs) emerged as the dominant approach for plant disease classification. Architectures such as AlexNet and VGGNet significantly improved performance by automatically learning hierarchical feature representations directly from raw leaf images without requiring manual feature engineering. These deep models achieved higher classification accuracy but required large labeled datasets and substantial computational resources for both training and inference [3], [4]. To overcome training limitations, transfer learning techniques were introduced. Pre-trained models such as ResNet and Inception, originally trained on large-scale datasets like ImageNet, were fine-tuned for plant disease classification tasks. This approach reduced training time and improved classification performance. However, these architectures often contained a large number of parameters, making deployment on edge devices and mobile platforms computationally expensive due to memory and processing constraints [5], [6]. EfficientNet introduced compound scaling techniques that systematically balance network depth, width, and input resolution. EfficientNet-based models demonstrated superior accuracy while maintaining lower parameter counts compared to conventional CNN architectures. Several studies applying EfficientNet variants for plant leaf classification reported improved accuracy along with computational efficiency, making them suitable for real-time agricultural applications [7], [8]. Multi-class disease detection systems have also been developed to classify multiple crops and disease categories simultaneously. These systems commonly utilized large datasets such as the PlantVillage dataset, which includes dozens of plant disease classes. Although classification accuracy improved significantly, many models lacked real-world validation under diverse environmental conditions, limiting their practical deployment in open agricultural fields [9], [10]. Data augmentation strategies such as rotation, flipping, scaling, and brightness adjustment have been widely implemented to enhance model robustness. These techniques improve the model's ability to generalize across varying leaf orientations and lighting conditions. However, excessive augmentation may introduce unrealistic distortions that do not accurately represent actual agricultural environments, potentially affecting prediction reliability [11], [12]. Beyond overall accuracy, recent research has emphasized the importance of evaluation metrics such as precision, recall, F1-score, and confusion matrix analysis. These metrics provide deeper insight into class-wise performance, especially in multi-class classification scenarios where class imbalance exists. Some studies reported high overall accuracy but showed lower recall values for minority disease classes, indicating possible misclassification risks [13], [14]. Statistical validation methods such as k-fold cross-validation have been adopted to improve reliability. In this approach, the dataset is divided into multiple folds, and the model is trained and validated across different subsets. Reporting the mean accuracy along with standard deviation enhances the credibility and robustness of the experimental results [15], [16]. Some recent systems have integrated environmental parameters such as temperature and humidity into plant disease detection frameworks. By incorporating live weather data through external APIs, these systems provide contextual irrigation and disease management recommendations. However, many implementations rely on static threshold-based rules rather than adaptive environmental modeling [17], [18]. Edge computing and lightweight deployment strategies have gained attention in smart agriculture applications. Model optimization techniques such as pruning and quantization reduce model size and inference time, making deployment feasible on resource-constrained devices. Although these techniques improve computational efficiency, they may introduce slight reductions in accuracy [19], [20]

## III. Proposed System

The proposed system is an Artificial Intelligence-based plant leaf disease detection and smart crop management platform designed to assist farmers and agricultural practitioners in identifying crop infections at an early stage and taking timely corrective action. The system integrates deep learning-based image classification with real-time environmental data analysis to provide accurate disease prediction, dynamic watering recommendations, and actionable agricultural guidance.

Unlike traditional manual inspection methods, the proposed framework provides an automated, reliable, and user-friendly decision-support tool. The architecture is modular, scalable, and computationally efficient. It focuses on clarity, real-time operation, and practical implementation through a web-based interface. This makes the system suitable for agricultural research, field experiments, and real-world farming applications.

### *I. System Architecture Overview*

The system follows a modular architecture where each component performs a specific task and communicates through defined interfaces. This modular design improves maintainability, scalability, and future extensibility.

The major modules include:

- Leaf Image Acquisition
- Image Preprocessing
- Disease Classification
- Remedy Recommendation
- Weather Data Acquisition
- Smart Watering Estimation
- Plant Health Assessment
- Visualization and User Interface

The system workflow begins with user interaction and ends with the generation of a comprehensive plant health report.

### *II. Leaf Image Acquisition Module*

This module serves as the entry point of the system. The user uploads a leaf image through the web interface. The system supports common image formats such as JPG and PNG, ensuring compatibility with both mobile and desktop devices.

Before processing, the uploaded image is validated to ensure acceptable resolution and clarity. This module allows non-technical users to interact with the platform easily without requiring prior expertise.

### *III. Image Preprocessing Module*

The image preprocessing module prepares the uploaded leaf image for classification. Each image is resized to 224 × 224 pixels to match the input requirement of the EfficientNet-B0 model. Pixel normalization and scaling are performed to reduce the effect of lighting variation, camera differences, and background noise.

Preprocessing ensures consistency across all inputs and improves overall classification reliability.

### *IV. Disease Classification Module*

The model is trained on labeled plant leaf datasets containing healthy and diseased samples across multiple crop types.

When an image is uploaded:

1. The preprocessed image is passed to the trained model.
2. The model extracts hierarchical visual features.
3. A probability score is computed for each disease class.
4. The class with the highest probability is selected as the final prediction.

5. The confidence percentage is displayed to the user.

The prediction uses the Softmax activation function:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}}$$

Where:

- $z_i$ is the output logit for class $i$
- $C$ is the total number of classes

*V. Remedy Recommendation Module*

After classification, the system retrieves disease-specific remedies from a structured knowledge base.

Each disease is associated with:

- Preventive measures
- Organic treatment suggestions
- Chemical treatment recommendations
- Crop management practices

This module converts technical classification results into practical agricultural guidance. If the plant is healthy, the system provides general maintenance recommendations instead of treatment.

*VI. Weather Data Acquisition Module*

Environmental conditions significantly influence plant health.

The system retrieves live weather data using external APIs such as:

- Temperature
- Humidity
- Atmospheric condition

These parameters are incorporated into the final report and influence watering recommendations.

*VII. Intelligent Watering Estimation Module*

This module determines the appropriate water requirement based on:

- Detected disease type
- Crop category
- Current temperature

A rule-based adjustment mechanism is used:

$$Water = BaseWater + (Temp - 25) \times 0.05$$

Fungal diseases reduce watering recommendations to prevent moisture accumulation.

*VIII. Plant Condition Assessment Module*

This module synthesizes:

- Disease classification
- Environmental conditions
- Watering requirements

The plant is categorized into:

- Healthy
- Diseased
- Water-stressed
- Low sunlight condition

An emoji-based representation enhances interpretability.

*IX. User Interface and Visualization Module*

The proposed system is built as an interactive web application using the Gradio framework. The interface leads users through a simple process: first uploading a leaf images, then entering their city to enable weather analysis, and finally starting the disease detection step. The result is shown in a clear and well-organized layout.

The output includes the predicted disease name, confidence score, suggested remedies current weather details, watering recommendations overall plant condition, and direct links to agricultural medicines available on online platforms such such as Amazon, Flipkart and eBay. The interface is designed to be intuitive, responsive and easy to use to use even for people with limited technical experience.

A. Dataset and Training Details

1. *Dataset Description:*

The dataset consists of publicly available plant leaf disease images including multiple crops and disease categories. Both healthy and diseased samples are included.

2. *Data Size and Split:*

- Training Data: 80%
- Testing Data: 20%

The model is trained on the training set and evaluated on the test set.

3. *Number of Classes:*

The model is trained to classify multiple plant disease categories including healthy leaves. Each class represents a specific plant disease condition or a healthy leaf category.

B. Performance Metrics

*Evaluation Metrics:*

The performance of the proposed system is evaluated using standard classification metrics such as Accuracy, Precision, Recall, and F1-Score. These metrics help in understanding the correctness and reliability of the model predictions.

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

F1 Score:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

A confusion matrix is also used to analyze class-wise performance and misclassification patterns.

C. Statistical Validation

1. *K-Fold Cross Validation:*

To validate the reliability of the model, K-Fold Cross Validation can be applied during training. In this method, the dataset is divided into K equal parts. The model is trained K times, each time using a different fold as validation data and the remaining folds as training data. The final performance is obtained by averaging the results.

2. *Standard Deviation of Accuracy:*

The standard deviation of accuracy helps measure the consistency of the model performance across different training runs or folds. A lower standard deviation indicates stable model performance.

Formula:

$$\sigma = \mathrm{F}\sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

3. *Statistical Reliability:*

Statistical reliability ensures that the model performance is consistent and not dependent on random variations in training data. Repeated experiments and validation techniques improve confidence in the results.

D. Complexity Analysis

1. Computational Complexity Analysis

- Model Parameter Count

EfficientNet-B0 contains approximately 5.3 million parameters, making it suitable for lightweight deployment.

- Model Size

The trained model occupies approximately 280 MB in .h5 format.

- Inference Time

The average inference time of the model is approximately 200–400 milliseconds depending on hardware and input size.

## IV. METHODOLOGY

a. Dataset Collection and Organization

The dataset used in this study consists of plant leaf images representing both healthy and diseased conditions across multiple crop types. Each image is carefully labeled according to its specific disease category or healthy class. The dataset is organized in a structured directory format, where each folder corresponds to one disease type or healthy category. This structured arrangement simplifies the training process and ensures that the model can easily differentiate between classes.

Special attention is given to maintaining a balanced number of samples for each category. Ensuring sufficient representation for every class helps prevent model bias and improves generalization performance during training.

b. Data Preprocessing and Augmentation

Before training and evaluation, all images undergo a standardized preprocessing pipeline. Each image is resized to 224 × 224 pixels to match the input requirements of the EfficientNet-B0 architecture. Pixel values are normalized and scaled to reduce the influence of lighting variations, camera differences, and background noise.

To enhance model robustness, data augmentation techniques are applied during training. These include small rotations, horizontal and vertical flipping, slight zooming, and controlled brightness adjustments. These transformations simulate real-world conditions and help the model learn meaningful patterns rather than memorizing the training dataset. As a result, the system performs more reliably when exposed to new leaf images.

c. Model Selection and Architecture Design

The proposed system employs a Convolutional Neural Network (CNN) based on the EfficientNet-B0 architecture. This model is selected because it balances accuracy and computational efficiency through a compound scaling strategy that optimizes network depth, width, and input resolution simultaneously.

To adapt the pre-trained EfficientNet-B0 model to the specific plant disease dataset, the final classification layer is modified to match the total number of disease categories. A Softmax activation function is applied at the output layer to generate probability scores for each class. The class with the highest probability is selected as the final prediction.

d. Model Training Strategy

The model is trained using labeled leaf images, allowing it to learn patterns associated with different disease categories. The dataset is divided into training and validation subsets to monitor performance and prevent overfitting. This separation ensures that the model does not simply memorize the training data but instead learns to generalize.

Categorical cross-entropy is used as the loss function because it is well-suited for multi-class classification tasks. The Adam optimizer is selected to adjust model weights efficiently during training. The training process runs for multiple epochs, and an early stopping mechanism is implemented to automatically halt training when validation performance no longer improves. This strategy prevents unnecessary computation and reduces overfitting.

e. Model Evaluation Metrics

To evaluate the effectiveness of the model, several performance metrics are used, including accuracy, precision, recall, and F1-score. These metrics collectively provide a comprehensive understanding of classification performance.

A confusion matrix is also generated to analyze class-wise predictions. It helps identify which disease categories are frequently misclassified, particularly in cases where visual similarities exist between diseases. This analysis provides insights for future dataset enhancement and model improvement.

f. Inference and Disease Prediction

During deployment, users upload a plant leaf image through the web interface. The image undergoes the same preprocessing steps used during training to ensure consistency. The processed image is then passed through the trained EfficientNet-B0 model.

The network computes probability scores for all possible disease categories. The class with the highest probability is selected as the predicted result. The corresponding confidence score is displayed to the user, enhancing transparency and trust in the system's prediction.

g. Remedy Mapping and Knowledge Integration

After detecting the disease, the system maps the predicted class to a structured remedy database. This database contains practical agricultural recommendations, including preventive measures, organic treatments, chemical treatment options, and general crop management practices.

In addition to textual guidance, the system may provide direct links to agricultural products available on platforms such as Amazon, Flipkart, and eBay. This integration ensures that the diagnostic process directly supports actionable steps for farmers.

h. Weather Data Processing

To incorporate environmental context, the system retrieves real-time weather information based on the user-provided city name. Weather parameters such as temperature, humidity, and atmospheric conditions are obtained through an external weather API.

This environmental information is displayed to the user and also contributes to refining watering recommendations, making the system more context-aware and location-specific.

i. Water Requirement Estimation Logic

The system uses a rule-based mechanism to estimate appropriate water requirements. The estimation considers the detected disease type, plant category, temperature, and humidity levels.

For example, when fungal diseases are detected or humidity levels are high, the system recommends reduced watering to prevent further disease spread. Conversely, during hot and dry conditions, increased watering may be suggested. The rule-based approach ensures transparency, interpretability, and adaptability across different agricultural scenarios.

j. Plant Condition Interpretation

Based on disease prediction, environmental conditions, and watering recommendations, the system summarizes the overall plant condition into intuitive categories such as Healthy, Stressed, Water-Deficient, or Low Sunlight.

This final interpretation step transforms complex analytical outputs into clear and understandable information, allowing users to make informed decisions without technical expertise.

## V. BLOCK DIAGRAM

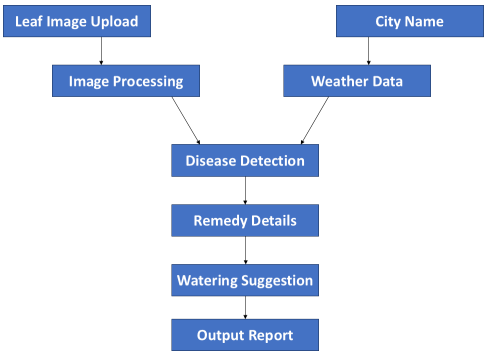


Fig. 1 AI-Based Plant Leaf Disease Detection System Architecture.

i. Leaf Image Upload

The process begins when the user uploads an image of a plant leaf using the application interface. The image can be captured using a mobile phone camera or selected from the device storage. This step allows users to provide real-time visual information about the plant condition. Care is taken to ensure that the uploaded image clearly shows the leaf surface, as this helps the system understand the plant's health condition more accurately.

ii. City Name Input

Along with the leaf image, the user is asked to enter the name of their city. This input is required to obtain local weather information related to the plant's environment. Since weather conditions vary from place to place, using the city name helps the system provide more relevant care guidance instead of giving generalized suggestions.

iii. Image Processing

After the image is uploaded, it is prepared for analysis. The image is adjusted to a uniform size and format so that it can be handled consistently by the system. This step helps reduce variations caused by different camera qualities, lighting conditions, or image resolutions. By standardizing the image, the system ensures smoother and more reliable analysis in the next stage.

iv. Disease Detection

Once the image is prepared, the system examines the leaf to understand its condition. At this stage, the system checks whether the plant appears healthy or shows signs of a disease. The identification is based on patterns observed in the leaf image, such as color changes or visible damage. The result of this stage determines the next set of actions taken by the system.

v. Disease Identification

After detecting the plant condition, the system assigns a clear label indicating whether the plant is healthy or affected by a specific disease. A confidence value is also associated with the result to indicate how reliable the identification is. This helps users trust the result and understand that the output is based on the clarity of the uploaded image.

vi. Remedy Suggestion

If the plant is identified as diseased, the system provides a list of simple remedy suggestions. These remedies are written in an easy-to-understand manner and focus on basic treatment and prevention steps. If the plant is found to be healthy, the system instead displays general care tips to help maintain the plant's good condition. This step ensures that users always receive useful guidance, regardless of the plant's health status.

vii. Weather Information

Using the city name provided earlier, the system retrieves current weather details such as temperature and humidity. These environmental factors play an important role in plant growth and care.

viii. Watering Recommendation

Based on the current weather conditions and the plant's health status, the system provides a watering recommendation. The suggestion indicates how much water the plant may require under the given conditions. This helps users avoid common issues such as overwatering or underwatering, which can negatively affect plant health.

ix. Result Display

In the final step, all the information generated by the system is presented to the user in a clear and organized format. The output includes the plant condition, disease name if detected, confidence

level, remedy or care suggestions, weather details, and watering guidance. The result is designed to be simple, readable, and actionable so that users can easily follow the recommended steps to take care of their plants.

## VII. Comparison of Existing Methods and Proposed

| **Existing Method / Approach** | **Limitations in Existing Systems** | **Proposed System Improvement** | **Advantage** |
|---|---|---|---|
| A deep learning model for plant disease detection using leaf images | System performs only classification and does not provide remedies or treatment guidance | The proposed system provides multiple remedies and care instructions along with prediction | Helps users take immediate corrective action |
| Image classification using conventional CNN models | Accuracy reduces when real-world leaf images are used | EfficientNet-based model improves feature extraction and accuracy | More reliable disease detection |
| Disease detection systems without environmental analysis | No watering or climate-based advices is given | Weather-based watering recommendations are included | Provides complete plant care guidance |
| Web-based plant diagnosis tools with limited functionality | No filtering of incorrect or unclear images, leading to false predictions | Low-confidence and non-leaf image rejection added | Improves reliability and correctness of results |
| Existing plant diagnosis systems with text-only output | User interface is not interactive and lacks visualization | Interactive UI with confidence bar, remedies, weather, and watering display | Better user experience and easier understanding |

Table. 1. Comparison of Existing Methods and Proposed System

## VI. Architectural Diagram

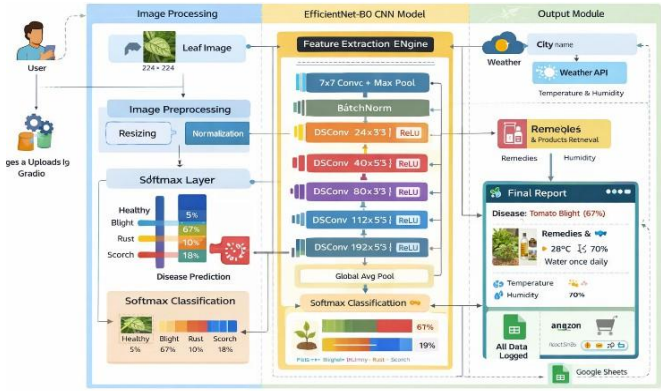


Fig. 2 The Architectural Origin of the Plant Care System

The Fig. 2s represents the complete workflow of the proposed AI Plant Doctor system, which detects plant leaf diseases and provides treatment and care recommendations using deep learning and environmental data. At the first stage, the user uploads a leaf image through the Gradio interface. The system receives the image and performs image preprocessing, which includes resizing the image to a fixed size (224×224 pixels) and normalization. These steps are required to make the image suitable for input into the deep learning model. After preprocessing, the image is passed to the EfficientNet-B0 Convolutional Neural Network (CNN). Which acts as the feature extraction engine. Inside the CNN, multiple convolutional layers extract important visual features such as color variations, texture patterns, and leaf spots. Batch normalization and activation functions improve training stability and model performance. Finally, a Global Average Pooling layer and Softmax classifier calculate the probability of each disease class and determine the final prediction.

Once the disease is identified, the system enters the output module. The detected disease is matched with a predefined knowledge base to retrieve recommended remedies and treatment steps. At the same time, the system retrieves real-time weather information (temperature and humidity) using an online weather API. Based on these environmental conditions, a watering recommendation is calculated to guide plant care.

## VIII. Results

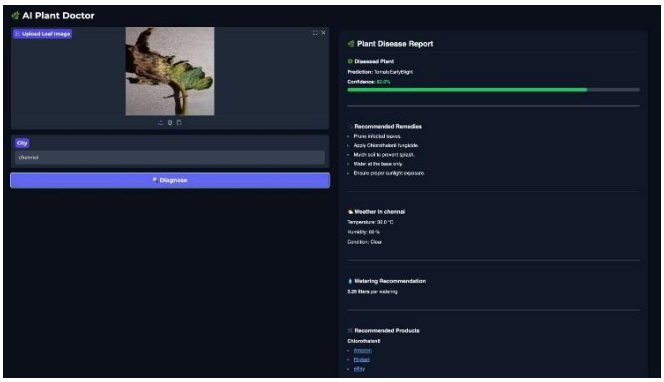


Fig. 3. Detection of Tomato Early Blight Leaf

In Fig. 3, a diseased tomato leaf image was uploaded to the AI Plant Doctor system. The model successfully detected the disease as Tomato Early Blight with a high confidence score. The system displayed recommended remedies such as pruning infected leaves

and applying fungicide. Weather details for the selected city were also retrieved, and a watering recommendation was generated based on temperature. Product links for treatment were displayed to assist in purchasing required fungicides.

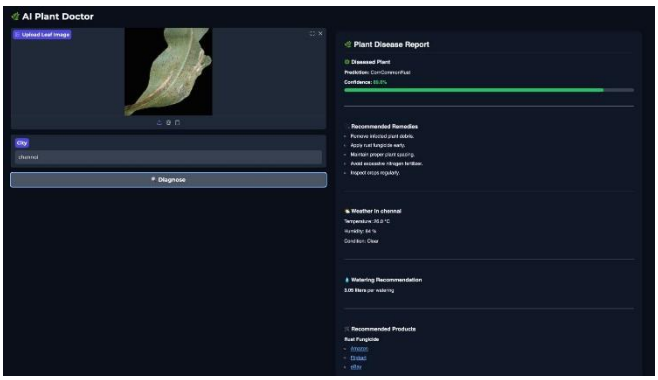


Fig. 4. Detection of Corn Common Rust Leaf

In Fig. 4, a corn leaf affected by rust disease was uploaded. The system correctly identified the disease as Corn Common Rust with high confidence. Multiple remedies were suggested, including removing infected debris and applying fungicide. The application also displayed weather conditions such as temperature and humidity, followed by a calculated watering recommendation. Relevant product links were provided to support treatment.

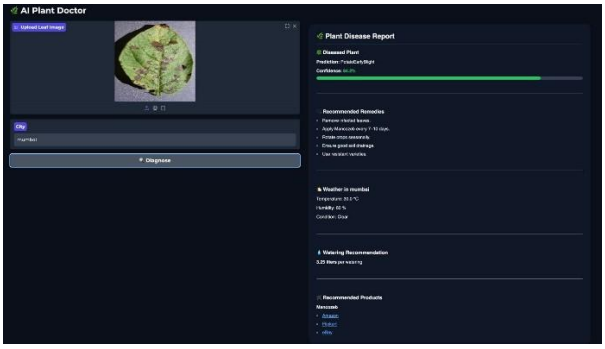


Fig. 5. Detection of Potato Early Blight Leaf

In Fig. 5, a potato leaf showing early blight symptoms was uploaded to the system. The model accurately predicted Potato Early Blight and displayed the confidence level. The system provided several remedies, including removing infected leaves and applying fungicide. Weather information for the selected city was shown, and watering quantity was recommended based on environmental conditions. Product links were also displayed to help users obtain the necessary treatment materials.

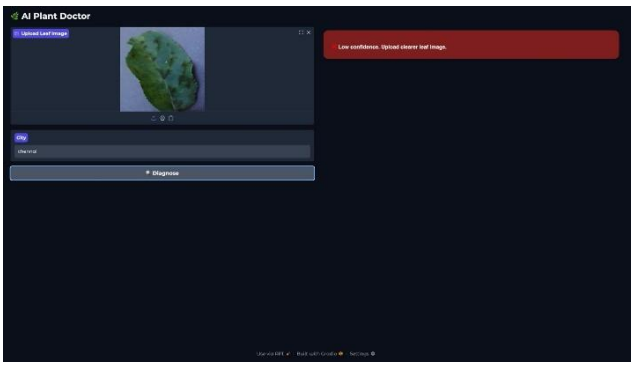


Fig. 6. System Response for Low-Quality or Unclear Leaf Image

Figure 6 shows the behavior of the AI Plant Doctor when an unclear or low-quality leaf image is uploaded. The uploaded leaf image does not contain sufficient visual detail for the model to confidently identify disease patterns such as spots, discoloration, or texture variations.

To prevent incorrect predictions, the system evaluates the model confidence score. When the confidence value falls below a predefined threshold, the system avoids generating a diagnosis and instead displays a warning message: “Low confidence. Upload clearer leaf image.”

This safety mechanism improves the reliability of the system by:

- Preventing false or misleading diagnoses.
- Encouraging users to upload high-resolution and properly focused leaf images.
- Ensuring that predictions are made only when the model confidence is adequate.

This feature is important in real-world agricultural applications where image clarity directly affects disease detection accuracy.

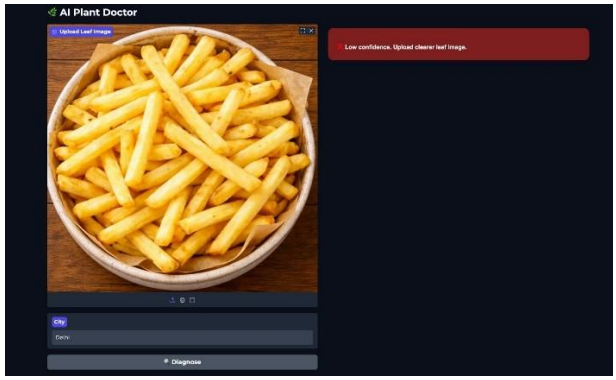


Fig. 7. System Response for Non-Plant Image Input

Figure 7 illustrates how the system responds when an uploaded image does not contain a plant leaf. In the example scenario, an image of food (French fries) is uploaded instead of a leaf image.

Before performing disease classification, the system executes a preliminary validation step. This step analyzes essential visual characteristics such as color distribution, texture patterns, and structural features to determine whether the uploaded image is likely to represent a plant leaf. This validation process prevents the model from directly classifying unrelated objects.

If the uploaded image does not satisfy the expected plant-leaf characteristics, or if the classification confidence is extremely low, the system does not proceed with disease prediction. Instead, it displays the following message to the user:

“Low confidence. Upload clearer leaf image.”

This validation mechanism ensures that:

- The model does not attempt to classify irrelevant objects.
- The overall system remains robust and reliable.
- Incorrect predictions caused by inappropriate inputs are minimized.

Such validation is essential in machine learning systems deployed in open environments, where users may accidentally or intentionally upload unrelated images. By incorporating this safeguard, the system maintains accuracy and practical usability.

## IX. CONCLUSION

The proposed project presents an intelligent plant leaf disease detection and plant care support system designed to assist users

in identifying plant health conditions through a simple and accessible web-based interface. The system requires only minimal inputs, specifically a leaf image and the user's city name, making it suitable for individuals without technical expertise.

Upon image upload, the system analyzes the leaf to determine whether the plant is healthy or affected by a disease. The diagnosis is presented clearly along with a confidence value, allowing users to understand the reliability of the prediction. This approach reduces dependence on manual visual inspection and enables faster identification of plant health issues.

Beyond disease detection, the system provides practical and easy-to-follow remedy suggestions. These recommendations focus on simple treatment measures and preventive care rather than complex agricultural procedures. If the plant is identified as healthy, the system offers general maintenance guidance, ensuring that the application remains useful regardless of the diagnosis.

To enhance real-world relevance, the system integrates live weather data based on the user-provided city. Environmental factors such as temperature and humidity are displayed as part of the result. These conditions are also used to generate watering recommendations, helping users avoid common mistakes such as overwatering or underwatering.

Overall, the system combines deep learning-based leaf image analysis with weather-aware care guidance to deliver a functional and user-friendly plant health monitoring solution. Its structured result presentation and clear recommendations make it suitable for home gardeners, students, and small-scale agricultural applications.

The modular design of the system allows for future enhancements, including support for additional plant species, expanded disease categories, and advanced care advisory features. This flexibility makes the proposed framework adaptable for future smart agriculture and intelligent plant care applications.

## REFERENCES


1. Sun, J., Li, X., & Zhang, H., "Research on plant diseaseidentification based on CNN (FL-EfficientNet)," *Computational Agriculture*, vol. 12, pp. 45–53, 2022.
2. Kanakala, R., & Ningappa, M., "Detection and Classification of Diseases in Multi-Crop Leaves using LSTMand CNN Models," *arXiv preprint*, arXiv:2505.00741, 2025.
3. Krishna, R., Sharma, P., & Gupta, A., "Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach," *Sensors*, vol. 8, no. 1, pp. 1–20, 2025.
4. Tan, M., Le, Q., "Plant leaf disease classification using EfficientNet," *Ecological Informatics*, vol. 60, pp. 101182, 2020.
5. Hamed, M., Ali, S., & Khan, T., "Plant Disease Detection Using Deep Learning (CNN/EfficientNetV2)," *International Journal of Intelligent Systems and Applications*, vol. 15, no. 6, pp. 23–35, 2023.
6. Veldandi, A., Prasad, S., & Kumar, R., "Identification of Plant Leaf Disease Using CNN and Image Processing," *Journal of Intelligent Plant Image Recognition Systems*, vol. 44, no. 1, pp. 10–18, 2024.
7. Li, W., Zhang, J.,& Zhao, X., "Rice & potato leaf disease classification using CNN," *ScienceDirect*, 2021.
8. Sharma, A., & Mehta, P., "Asurvey ofdeep CNN models for plant leaf disease," *International Journal of Advanced Research in Science & Engineering*, vol. 8, no. 5, pp. 12–25, 2021.
9. Gupta, R., & Singh, V., "Deep CNN survey for plant leaf disease prediction," *PubMed*, 2021.
10. Kumar, A., & Joshi, S., "Plant leaf disease detection & classification using CNN models," *Springer*, 2025.
11. Zhao, H., Li, X.,& Chen, J., "Mobile-friendly deep learning across 101 classes for plant leaf disease detection," *arXiv preprint*, arXiv:2508.10817, 2025.
12. Patel, S., & Reddy, K., "Interpretable plant leafdiseasedetection using attention-enhanced CNN," *arXiv preprint*, arXiv:2512.17817, 2025.
13. Sharma, R., & Verma, D., "EfficientNet-based plant disease classification with ~95% accuracy," *ResearchGate*, 2025.
14. Sun, Y., Li, X., & Huang, J., "Combining ML and DL for diverseleaf disease datasets," *PubMed*, 2025.
15. Khan, R., & Ali, S., "Guava fruit & leaf disease detection with quantized CNNs," *arXiv preprint*, arXiv:2209.12557, 2022.
16. Mohanty, S. P., Hughes, D. P., & Salathé, M., "Using Deep Learning for Image-Based Plant Disease Detection," *Frontiers in Plant Science*, vol. 7, pp. 1419, 2016.
17. Too, E. C., Yujian, L., Njuki, S., & Yingchun, L., "A Comparative Study of Fine-Tuning Deep Learning Models for Plant Disease Identification," *Computers and Electronics in Agriculture*, vol. 161, pp. 272–279, 2019.
18. Ferentinos, K. P., "Deep Learning Models for Plant Disease Detection and Diagnosis," *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.
19. Ramcharan, A., McCloskey, P., Baranowski, K., et al., "A Mobile-Based Deep Learning Approach for Cassava Disease Detection," *Frontiers in Plant Science*, vol. 8, pp. 1852, 2017.
20. Tan, M., & Le, Q., "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," *Proceedings of the International Conference on Machine Learning (ICML)*, 20